Outline

1 Introduction

2 Clustering

3 Examples of clustering algorithms

4 Open problems in clustering

5 Bibliography

Picking a similarity measure

Whatever the type of clustering algorithm, they all rely on a similarity measure. Picking a good similarity measure is therefore of paramount importance.

- It is the core of any model.
- It determines what type of structures are considered interesting.
- It will ultimately change the clustering result.

Remarks

- Choosing a similarity measure, or building one often require a good knowledge of the data to analyze, or at least of the field they come from.
- Picking a similarity measure already introduces a bias in a task that is supposed to be exploratory.

Picking a similarity measure

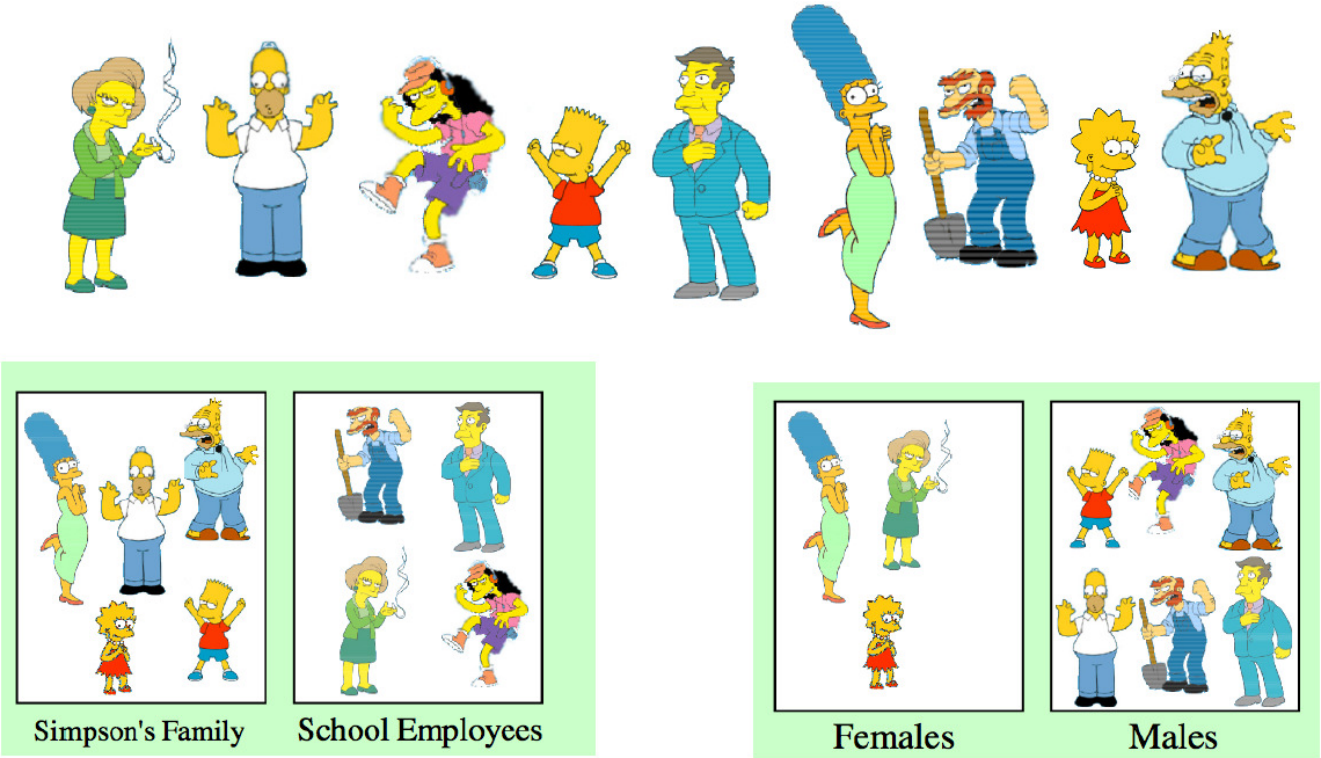


Figure: Depending on the similarity measure, both partitions can make sense.

Validating clustering results

Validating clustering results is a difficult process:

- There is no proper definition of what a "good cluster" is, or looks like.
- There is no "ground truth" in unsupervised learning to check the quality of a partition.
- Clusters come in all shapes and forms.
- The number of clusters to be found is usually unknown, and sometimes there are none.
- Not all data sets have well-defined clusters.

Identifying clusters

While it is usually obvious to visually spot the clusters in 2D or 3D data sets, it is almost impossible to do so when there are more features !

Internal indexes

Internal indexes *are criteria that can be used to evaluate the quality of a clustering partition.*

- Internal indexes can be subjective as they favor some cluster shapes, and can be biased toward a lower number of clusters.
- They usually assess the compactness of the clusters and whether they are well separated.

Remark

Internal indexes are said to be "internal", by opposition to "external indexes" that compare a result with an external ground truth.

Internal indexes: Davies-Bouldin Index

Let S_i be the average scatter of a cluster c_i around its mean value μ_i:

$$S_i = \frac{1}{|c_i|} \sum_{x \in c_i} ||x - \mu_i||_2$$

Let $M_{i,j} = ||\mu_i - \mu_j||_2$ be the average distance between two clusters c_i and c_j.

Davies-Bouldin Index for K clusters

$$DB = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \frac{S_i + S_j}{M_{i,j}}$$

Internal indexes: Davies-Bouldin Index

The Davies-Bouldin index has the following properties:

- A lower DB value means a better clustering.
- This index is not normalized.
- It favors spherical clusters.
- it is biased so that it gives lower values with less clusters.

Remark

The Davies-Bouldin index is usually the favored internal index used to evaluate partitions created using the K-Means algorithm.

Internal indexes: Silhouette Index

The Silhouette index is another internal index with interesting properties:

- It is normalized between -1 and 1 (a value bellow 0 meaning a bad partition).
- It can evaluate if a a data belongs to a cluster, if a cluster is well formed, or the whole partition.

Let $a_x \simeq ||x - \mu_i||_2$ be the mean distance between $x \in c_i$ and the data that belong to the same cluster. And b_x the mean distance to the data that belong to the closest other clusters.

$$b_x = \min_{k \neq i} \frac{1}{|c_k|} \sum_{y \in c_k} ||x - y||_2$$

Silhouette index of an element x

$$SC(x \in c_i) = \frac{b_x - a_x}{max(a_x, b_x)}$$

Internal indexes: Silhouette Index

Silhouette index of a cluster c_i

$$SC(c_i) = \frac{1}{|c_i|} \sum_{x \in c_i} SC(x)$$

Silhouette index of a partition

$$SC = \frac{1}{K} \sum_{i=1}^{K} SC(c_i)$$

- The silhouette index best value is 1.
- It favors spherical clusters.

Range	Interpretation
$S > 0.70$	Strong structures have been found
$0.5 < S < 0.70$	reasonable structures have been found
$0.25 < S < 0.5$	Weak structures have been found
$0 < S < 0.25$	No substantial structure found
$S < 0$	This clustering is most likely garbage

Table: Interpreting Silhouette values

Internal indexes: Calinski & Harabasz

This index is defined as follows:

$$CH(k) = \frac{B/(k-1)}{W/(N-k)}$$

- B is the sum of square distances between clusters
- W is the sum of square distances within clusters
- k is the number of clusters

Properties of the CH-Index

- Not normalized
- Better when higher
- With balanced clusters, the CH index is generally a good criterion to indicate the correct number of clusters.

Using External indexes to asses cluster quality

Quite often clustering algorithms will be tested in a non-exploratory setting:

- A data set for which the real classes are known will be used for test purposes (with the added advantage that the number of clusters will be known)
- In this case, the cluster found will be compared to the real classes using external indexes or purity measures.

Cluster purity

Given K real classes $Y^1, \cdots, Y^K$:

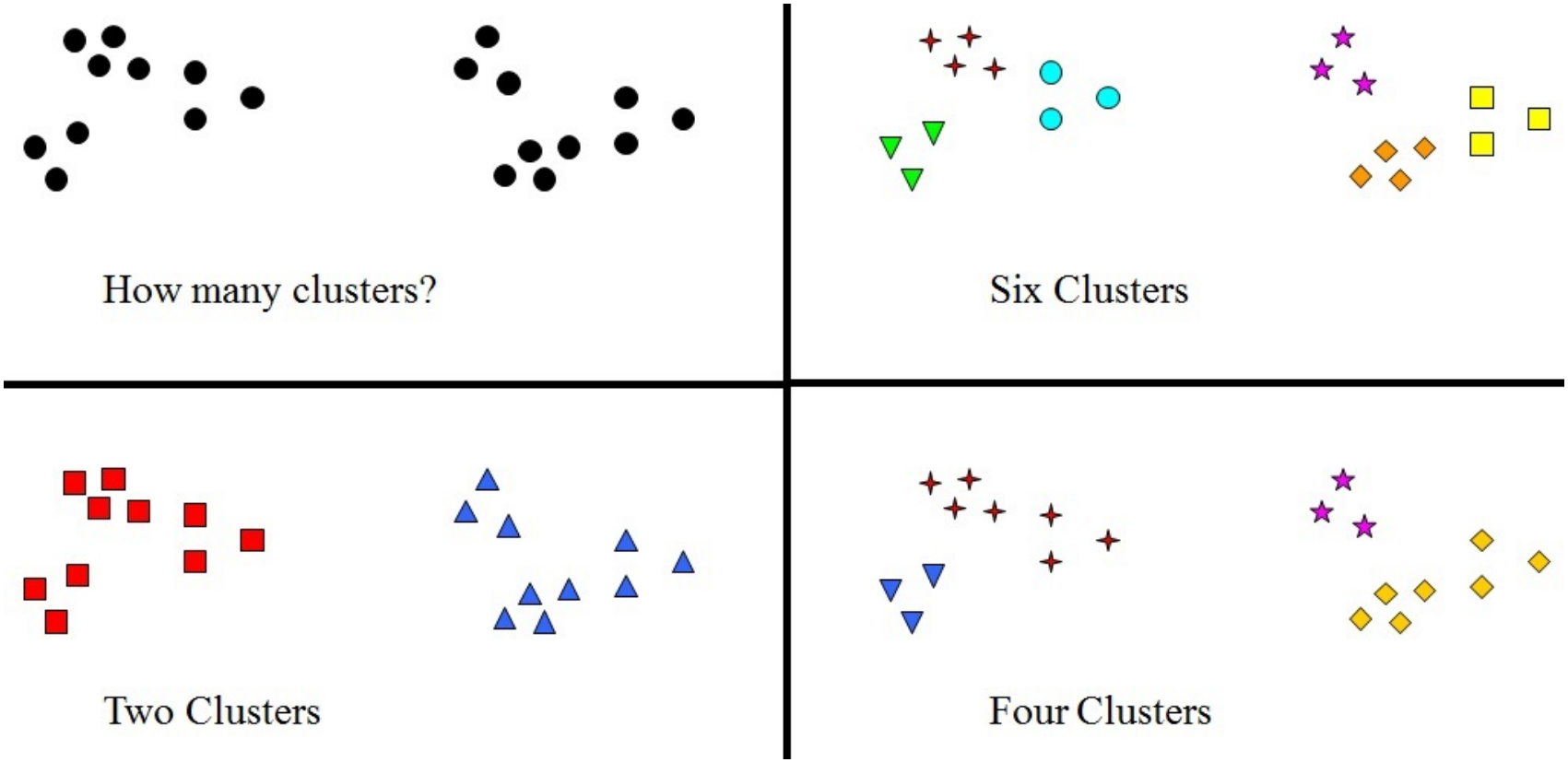
- Purity of a cluster C^i: $CP(C^i) = \frac{1}{|C^i|} argmax_j |C^i \cap Y^j|$
- Purity of a partition: $CP = \frac{1}{N} \sum_{i=1}^{k} argmax_j |C^i \cap Y^j|$

Examples of external indexes

- Rand Index, Adjusted Rand Index, Accuracy, etc.

How many clusters ?

Guessing the right number of clusters is one of the oldest problem in clustering, and several methods require it as a parameter.



Criteria to pick the right number of clusters

Some criteria exist to pick the optimal number of clusters, but they require to run the algorithm several times with different number of clusters in order to evaluate the solutions:

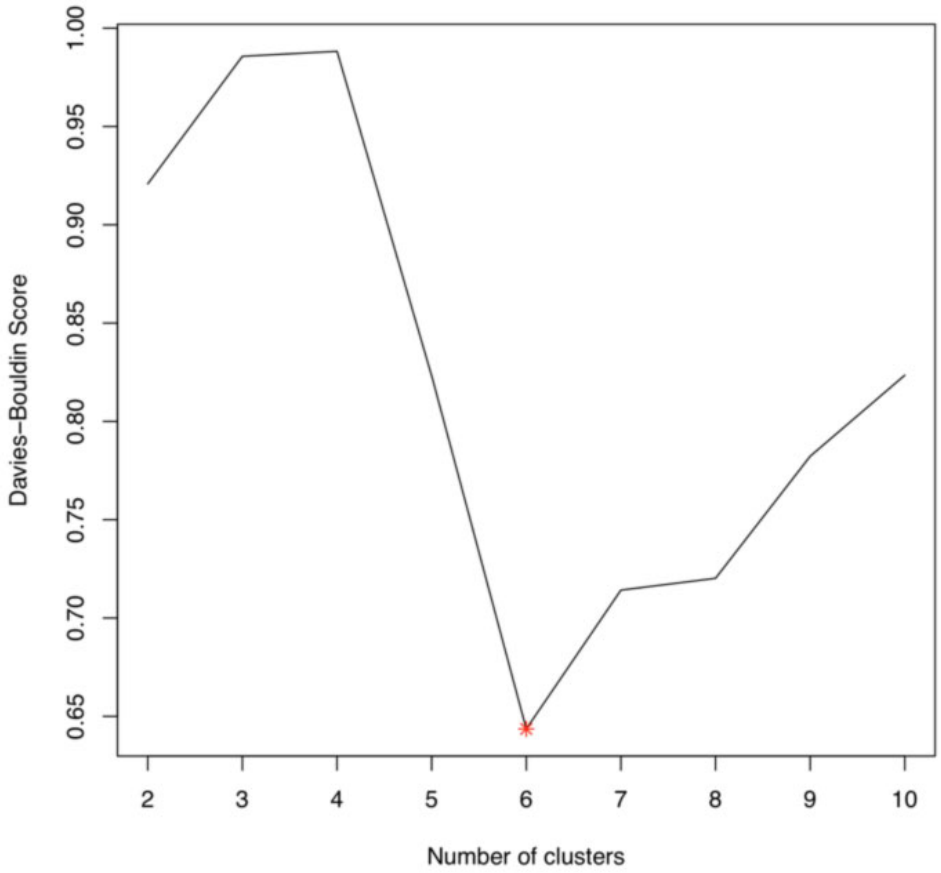
- The **Bayes Information Criterion** (BIC) and **Akaike Information Criterion** (AIC) work well with probabilistic algorithms.
- The **Minimum Description Length** criterion (MDL) and the **Minimum Message Length** criterion (MML) are recommended for hierarchical clustering.
- Stability can also be used to guess the right number of clusters.

Remark

Most of the time, the best solution is to ask the advice of an expert in the field of the studied data.

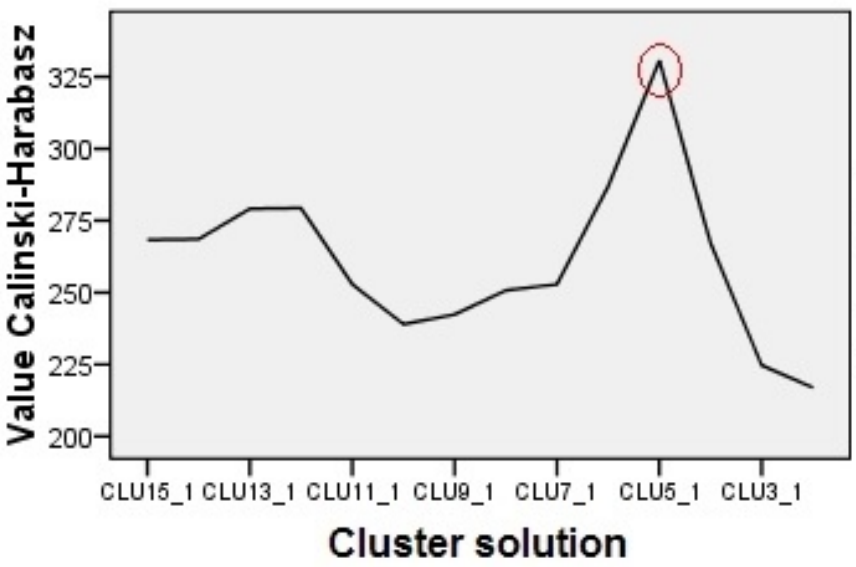
Criteria to pick the right number of clusters

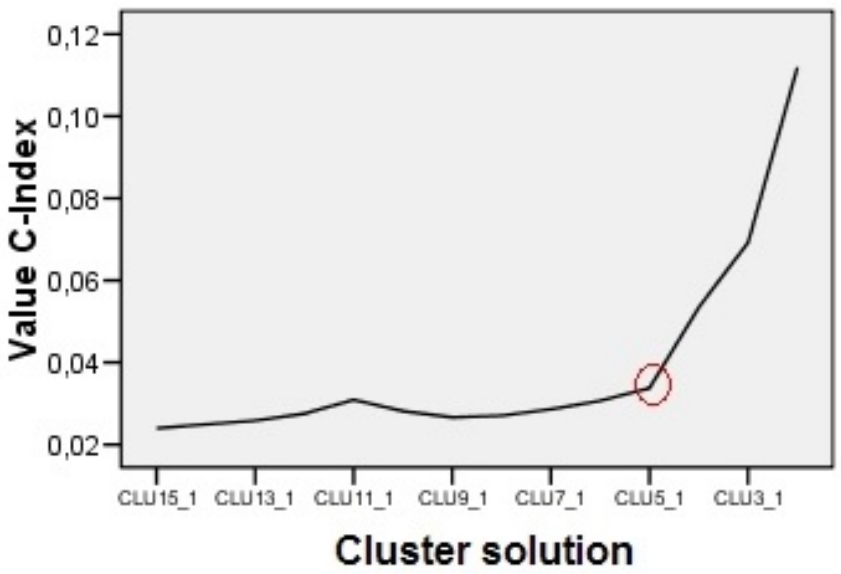
Sometimes clustering indexes can be used to help picking the right number of clusters:



Criteria to pick the right number of clusters

Another example with the C-Index and the Calinski-Harabasz indexes:





Picking a clustering algorithm

Picking the right clustering algorithm depends on several factors and requires prior knowledge on the data. With high dimensional data sets, you usually don't have this knowledge:

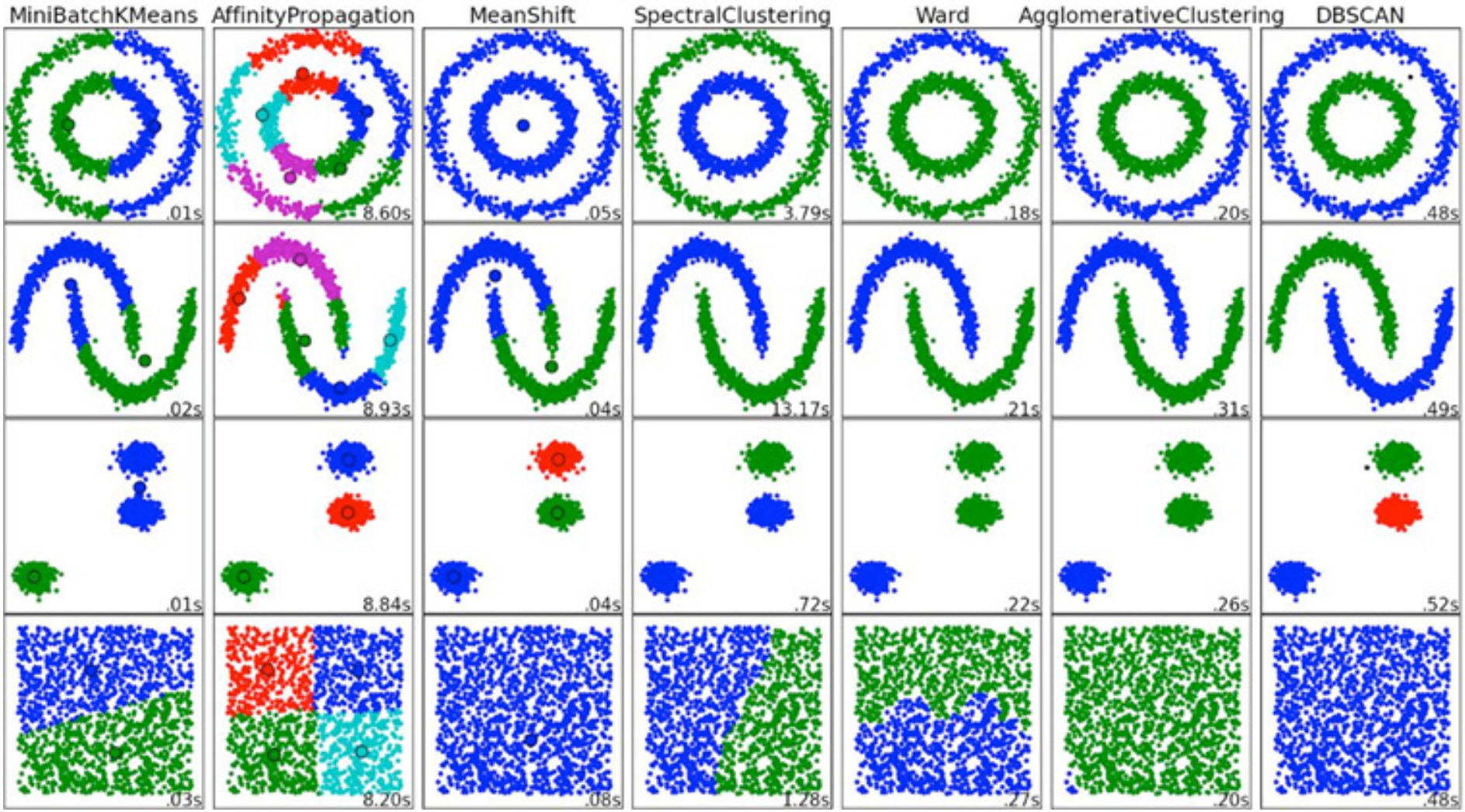
- Are there any clusters ?
- What is the shape of the clusters and how well separated are they ?

Other considerations may include the complexity of your algorithm:

- Complexity depending on the size of the data set.
- Complexity depending on the number of attributes of the data set.

Picking an algorithm often ends up being a trade-off between the expected quality of the result and the computational complexity of your algorithm.

Picking a clustering algorithm



Summary of open problems in clustering

- Defining the notion of similarity, and picking or defining a distance function accordingly.
- Picking the right algorithm and the right parameters for this algorithm.
- Figuring out how many clusters to search for.
- Evaluating clustering results.
- Comparing clustering results.
- All known data analysis problems: normalizing the data or not, picking relevant attributes, removing the outliers or not, etc.

Outline

1 Introduction

2 Clustering

3 Examples of clustering algorithms

4 Open problems in clustering

5 Bibliography

Bibliography

- Christopher M. Bishop, Pattern Recognition and Machine Learning (2006)
- Brian S. Everitt et al., Cluster Analysis 5th edition (2011)



Questions?